Rotation Matrices Part 1

EGR 445/545

GVSU

Notes

- clean up Rz derivation stuff
  - move some to review
  - get rid of some
- add in stuff about multiple rotations
- and HT preview and side view content to LA 1 review
- you also need some content on finding rotation matrices by inspection
  - I guess there is some

Plan for Tonight

- review:
  - LA 01
  - R_z derivation
    - columns are unit vectors
- R_y derivation (you)
- multiple rotations
- rotation matrices by inspection
  - multiple/compound rotations
- the transpose is the inverse of a rotation matrix
- learning activities 2 and 3

Thanks

- thanks for a good first class
- I think we are off to a solid start
  - everyone has jupyter running and my robotics module installed (I think)
- I commend you
  - this is not an easy course, but you chose it

Quick Review

- what did we learn last time?
  - what is the primary challenge in industrial robotics?
  - why do we need to learn robot kinematics?
  - what is a rotation matrix and where do they come from?

LA 1 Review: Side view

- what can we learn from a side view?
- what frame is it in?

LA 1 Review: Top view

- how does a top view help us get to a trig answer check for ${}^{0}P_{tip}$?

LA 1: Intro to HT Matrices

for ${}^{2}P_{tip}$, most people did this:

$$ {}^{2}P_{tip} = {}^{2}_{3}\mathbf{R} \cdot {}^{3}P_{tip} + \begin{bmatrix} L_1 \\ 0 \\ 0 \end{bmatrix} $$

but can we do this in one operation?

LA 1: DH Preview

How could we make all of the θ_i rotations about Z axes?

- **hint:** Z_1 and Z_2 are not parallel, so we will need an intermediate frame

Quick Segue: Multiple Rotations

- let the notation guide you
- more later

Review of Rotation Matrices

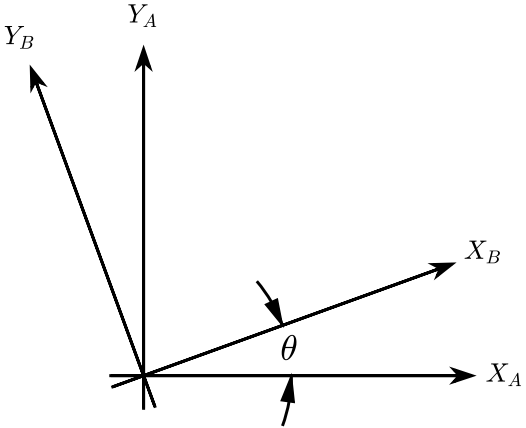
Understanding Rotation Matrices

What does this equation mean?:

$$
{}^{A}_{B}\mathbf{R} = \begin{bmatrix} {}^{A}\hat{X}_B & {}^{A}\hat{Y}_B & {}^{A}\hat{Z}_B \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}
$$

Rotation Matrix Sign Convention

- we rotate from A to B, but map from B to A
  - start with frames A and B aligned/coincident
  - rotate frame B away from A in a right-hand sense
  - ${}^{A}_{B}\mathbf{R}$ maps a vector from frame B back to frame A



Signs are important

- you have a growing suspicion that memorizing or googling rotation matrix is very risky as far as signs go
  - the same applies when using them correctly
- getting the signs right can be tricky when it comes to rotation matrices
- if you understand where rotation matrices come from, you will be protected from these sign errors

Signs are important - continued

Ginsberg:

$$[R_x] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & \sin\theta_x \\ 0 & -\sin\theta_x & \cos\theta_x \end{bmatrix}$$

$$[R_y] = \begin{bmatrix} \cos\theta_y & 0 & -\sin\theta_y \\ 0 & 1 & 0 \\ \sin\theta_y & 0 & \cos\theta_y \end{bmatrix}$$

$$[R_z] = \begin{bmatrix} \cos\theta_z & \sin\theta_z & 0 \\ -\sin\theta_z & \cos\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Craig:

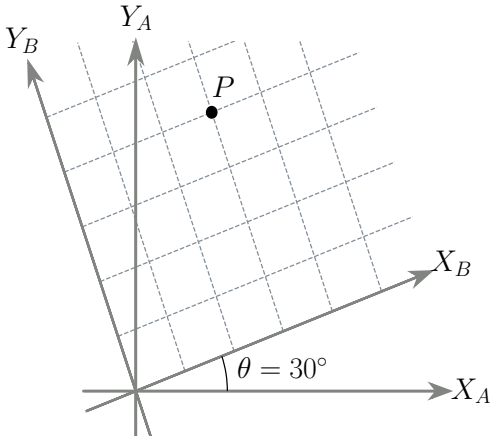
$$R_X(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

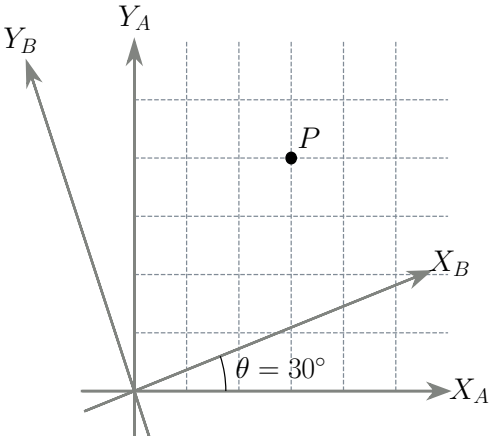
$$R_Y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_Z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

These are two different problems

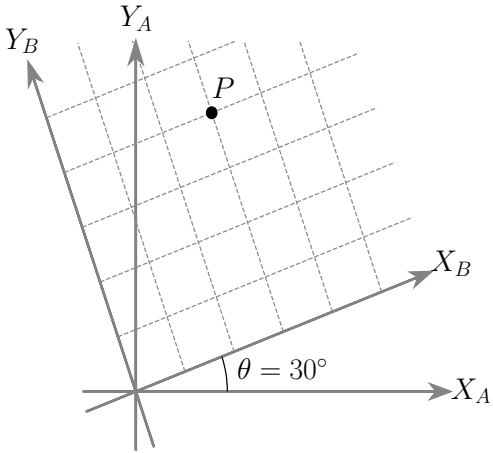
- what's the difference?

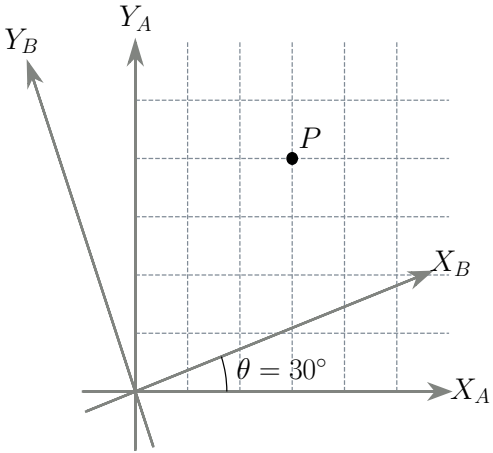




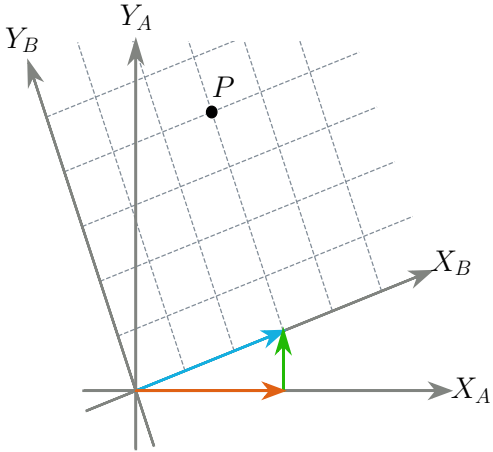
Which is which?

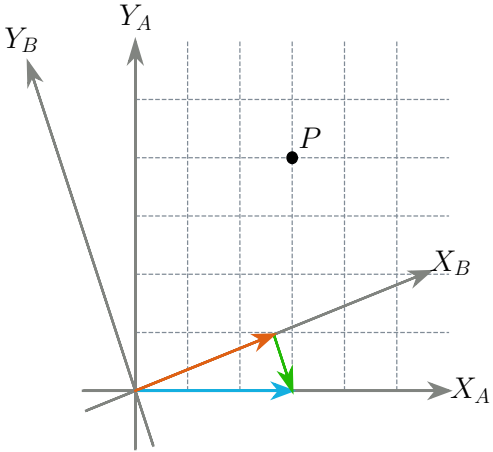
$$ {}^{B}P = {}^{B}_{A}\mathbf{R} \cdot {}^{A}P \quad \text{vs.} \quad {}^{A}P = {}^{A}_{B}\mathbf{R} \cdot {}^{B}P $$





What are the signs of the components in the other frame?





Review Exercise: Derive R_y

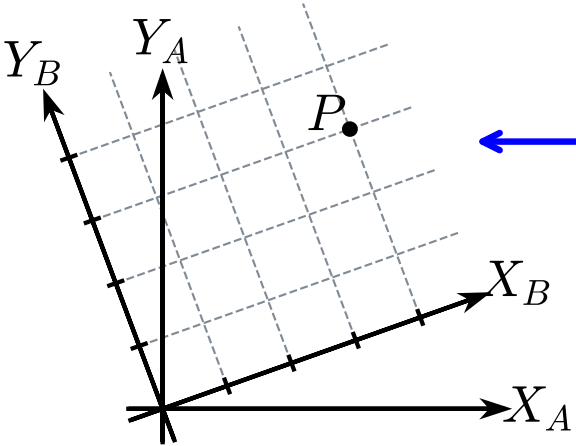
- draw a sketch and come up with an expression for ${}^{A}_{B}\mathbf{R}$ for a positive rotation about the Y axis

Conventions for rotations and mappings

- start with frames A and B coincident
- frame B rotates by angle θ
  - we rotate from A to B
  - follow the right hand rule
- ${}^{A}_{B}\mathbf{R}$ maps from frame B back to frame A
- this might seem weird at first
- it corresponds to what we know and what we want to find
  - when $\theta = 0$, frames A and B are aligned
  - when we actuate the joint, frame B moves
  - we ultimately know P_{tip} in the last frame and want to map back to frame 0

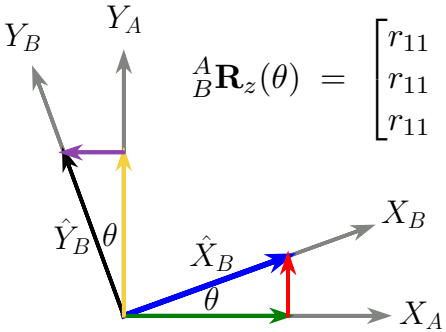
R_z Derivation Review

Rotation Matrix Intro



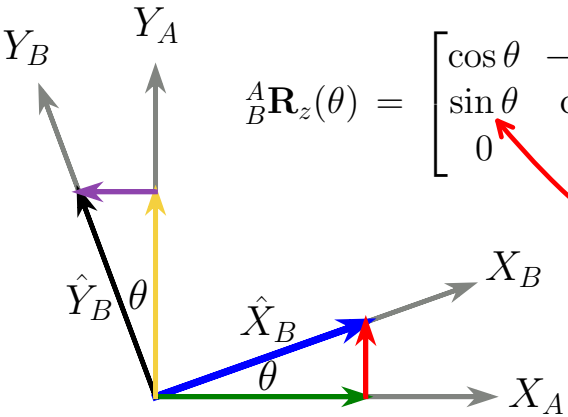
$$^{A}P = {}^{A}_{B}R\,{}^{B}P$$

Rz Derivation Overview



$$ {}^{A}_{B}\mathbf{R}_z(\theta) = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{11} & r_{12} & r_{13} \\ r_{11} & r_{12} & r_{13} \end{bmatrix} = \begin{bmatrix} {}^{A}\hat{X}_B & {}^{A}\hat{Y}_B & {}^{A}\hat{Z}_B \end{bmatrix} $$

R_z Derivation Summary



$$\hat{Y}_B \cdot \hat{X}_A$$

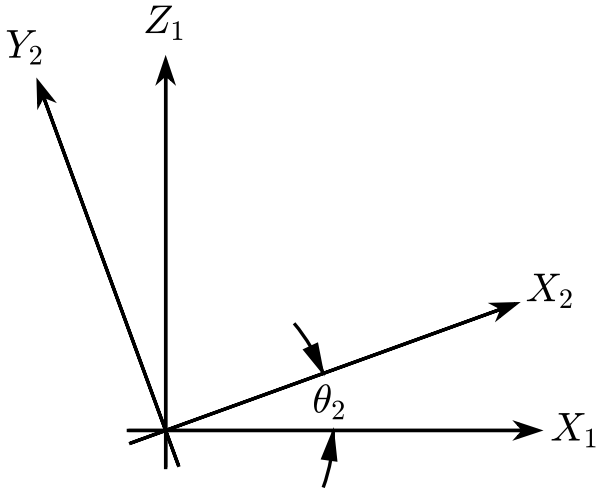
$$ {}^{A}_{B}\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

$$\hat{X}_B \cdot \hat{Y}_A$$

Multiple Rotations

Multiple Rotations

- why can't we easily find $\frac{1}{2}\mathbf{R}$ with one rotation?



Multiple Rotations

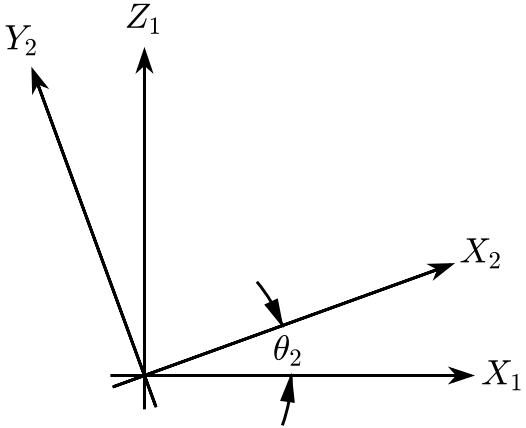
- if it takes two rotations to get from frame A to frame B, define an intermediate frame I
- first rotate from frame A to frame I
- then rotate from frame I to frame B
- the combined rotation matrix will be

$$ {}^{A}_{B}\mathbf{R} = {}^{A}_{I}\mathbf{R} \cdot {}^{I}_{B}\mathbf{R} $$

Multiple Rotations: ${}^1_2\mathbf{R}$ for the Plexiglass Robot

Do the θ_2 rotation second

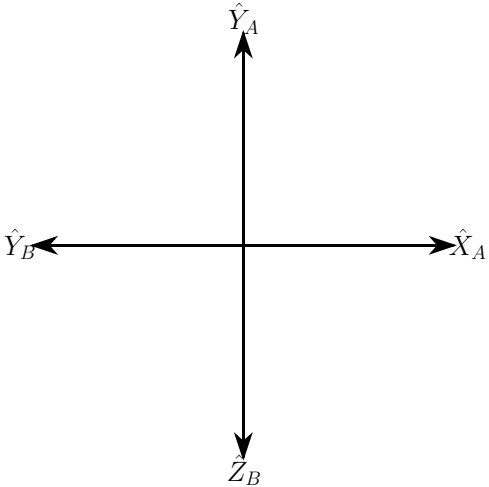
- we want θ_2 to rotate about $\hat{Z}_2$
- what would the first rotation be?
  - how do we get $\hat{Z}_2$ pointing where we need?
- where would the intermediate axes point?



Rotation Matrices by Inspection

Finding ${}^{A}_{B}\mathbf{R}$ with Arbitrary Orientation

What is ${}^{A}_{B}\mathbf{R}$ for this sketch?



The Inverse of a Rotation Matrix

Dot Products and Rotation Matrices

What does this mean? (eqn 2.3):

$$
{}^{A}_{B}\mathbf{R} = \begin{bmatrix} {}^{A}\hat{X}_B & {}^{A}\hat{Y}_B & {}^{A}\hat{Z}_B \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}
$$

The Transpose is the Inverse

- what do we know about the columns of ${}^{A}_{B}\mathbf{R}$?
  - what does this tell us about ${}^{A}_{B}\mathbf{R}^{T}$?

Dot Products Part 2

$$ {}^{A}_{B}\mathbf{R} = \begin{bmatrix} {}^{A}\hat{X}_B & {}^{A}\hat{Y}_B & {}^{A}\hat{Z}_B \end{bmatrix} \text{ (unit vectors as columns)} $$

$$ {}^{A}_{B}\mathbf{R}^T = \begin{bmatrix} {}^{A}\hat{X}_B^T \\ {}^{A}\hat{Y}_B^T \\ {}^{A}\hat{Z}_B \end{bmatrix} \text{ (unit vectors as rows)} $$

$$ {}^{A}_{B}\mathbf{R}^T \cdot {}^{A}_{B}\mathbf{R} = ? $$

Transpose as Inverse Conclusion

$$ {}^{A}_{B}\mathbf{R}^{T} = \begin{bmatrix} {}^{A}\hat{X}_{B}^{T} \\ {}^{A}\hat{Y}_{B}^{T} \\ {}^{A}\hat{Z}_{B} \end{bmatrix} \quad \text{and} \quad {}^{A}_{B}\mathbf{R} = \begin{bmatrix} {}^{A}\hat{X}_{B} & {}^{A}\hat{Y}_{B} & {}^{A}\hat{Z}_{B} \end{bmatrix} $$

$$ {}^{A}_{B}\mathbf{R}^{T} \cdot {}^{A}_{B}\mathbf{R} = \begin{bmatrix} {}^{A}\hat{X}_{B}^{T} \cdot {}^{A}\hat{X}_{B} & {}^{A}\hat{X}_{B}^{T} \cdot {}^{A}\hat{Y}_{B} & {}^{A}\hat{X}_{B}^{T} \cdot {}^{A}\hat{Z}_{B} \\ {}^{A}\hat{Y}_{B}^{T} \cdot {}^{A}\hat{X}_{B} & {}^{A}\hat{Y}_{B}^{T} \cdot {}^{A}\hat{Y}_{B} & {}^{A}\hat{Y}_{B}^{T} \cdot {}^{A}\hat{Z}_{B} \\ {}^{A}\hat{Z}_{B}^{T} \cdot {}^{A}\hat{X}_{B} & {}^{A}\hat{Z}_{B}^{T} \cdot {}^{A}\hat{Y}_{B} & {}^{A}\hat{Z}_{B}^{T} \cdot {}^{A}\hat{Z}_{B} \end{bmatrix} $$

$$ {}^{A}_{B}\mathbf{R}^{T} \cdot {}^{A}_{B}\mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

Learning Activity 2

- practice with rotation matrices

Learning Activity 3

Numeric verification of

$$ {}^{B}_{A}\mathbf{R} = {}^{A}_{B}\mathbf{R}^{T} = {}^{A}_{B}\mathbf{R}^{-1} $$